

SENIOR FULL-STACK ENGINEER - TAKE-HOME EXERCISE

Thank you for your interest in the Senior Full-stack Engineer role. As the next step in our process, we would like to invite you to complete a take-home exercise. It is designed to reflect real work rather than puzzle solving, and we are more interested in your judgment, assumptions and trade-offs than in feature volume.

Expected effort: 6 hours maximum in total

Submission deadline: Within 5 calendar days of receiving this brief

Follow-up session: 60 to 90 minutes with the CTO, where you will walk through and extend your work

Part 1: Build a Feature Slice

Build a small habit-tracking service:

- **Backend (Node.js):** An API where users can log daily health habits (for example: sleep, exercise, water intake) and retrieve their progress.
- **Frontend (React):** A simple view showing a user's current streaks and weekly progress. Functional and clear beats beautiful.
- **Database:** Use MySQL or MongoDB. Briefly justify your choice in the README.

The requirements are intentionally light. Where the brief is ambiguous, make a sensible decision and document it. Your README should cover:

- Your assumptions and the edge cases you considered (and which ones you deliberately did not handle)
- How to run the project locally
- What you would do differently for production, and where you cut corners given the time box

We do not expect production perfection in 6 hours. We expect you to know the difference and be explicit about it.

Part 2: Code Review

Alongside this brief, you will receive a short code sample (a Node.js API excerpt with a small React component). Treat it as a colleague's pull request and write your review as you would in a real team: identify the issues you see, explain why they matter, prioritise them, and suggest fixes. Written comments in a markdown file are perfect.

AI Tools

You are encouraged to use AI coding tools (Copilot, Claude Code, Cursor, or any other) in any part of this exercise, exactly as you would on the job. In your README, briefly tell us:

- Which tools you used and for what
- Where they helped, and where you overrode or corrected them

There is no penalty for using AI extensively or for using none at all. What we care about is that you fully understand and own everything you submit.

Ground Rules

- Please respect the 6 hour time box. We will ask how you spent your time.
- Submit your work as a Git repository (link or archive) containing the project, your README, and your Part 2 review file.
- This is a fictional exercise created for hiring. Your work remains yours and will not be used commercially.

The Follow-up Session

In the follow-up session you will walk us through your solution and reasoning, discuss your code review findings, and make a small live extension to your own code together with the team. Nothing adversarial; it is the kind of technical conversation we have every day.

If you have any questions or need a short extension due to your current commitments, please reach out. We look forward to seeing your work.